Modern Pace Handicapping: A Synthesis for The Analyst

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1. Introduction: Why Pace is the Key to the Kingdom

As you correctly identified after the All-Star Mile, final time is a lie. A horse that runs 1:36.00 after setting a blistering early pace has run a fundamentally different (and better) race than a horse that runs the same time after a slow, uncontested lead. The central thesis of Tom Brohamer's *Modern Pace Handicapping* is that **it's not how fast a horse runs, but *how* a horse runs fast.** [1]

This document synthesizes the core principles of Brohamer's methodology from a variety of publicly available sources, including articles from Brisnet, TrackMaster, and other publications that have covered his work. It is designed to give you the complete intellectual framework of the book, even without the original text.

2. The Four Pillars of Brohamer's Methodology

Brohamer's system is built on four key concepts that work together to create a complete picture of a race's pace shape and a horse's energy distribution. These are:

1. **Velocity-Based Pace Figures (Feet per Second):** The foundational measurement.
2. **The Track Decision Model:** Understanding the pace profile of the specific track and distance.
3. **Percent Early (%E):** Quantifying a horse's energy expenditure.
4. **Turn Time (The Hidden Fraction):** The single most important indicator of a horse's form and fitness.

We will cover each in detail.

3. Pillar 1: Velocity-Based Pace Figures (Feet per Second)

The entire system is built on converting times and lengths into a universal measurement: **feet per second (fps).** This allows for direct, objective comparison of horses across different distances and even different races. [2]

3.1. The Internal Fractions

Brohamer divides every race into three internal fractions:

Race Type	Fraction 1 (Fr1)	Fraction 2 (Fr2) - Turn Time	Fraction 3 (Fr3)
Sprints	Gate to 2 furlongs	2 furlongs to 4 furlongs	4 furlongs to finish
Routes	Gate to 4 furlongs	4 furlongs to 6 furlongs	6 furlongs to finish

Source: [3]

3.2. The Calculation

The calculation for each fraction is simple:

> ***Velocity (fps) = Distance of Fraction (in feet) / Time of Fraction (in seconds)***

For example, in a 6-furlong sprint (3,960 feet), a horse runs the second fraction (from the 2f pole to the 4f pole, a distance of 1,320 feet) in 23.1 seconds. Its velocity for that fraction is:

> *1320 feet / 23.1 seconds = **57.14 fps***

This number is the horse's **Turn Time** velocity.

3.3. The Key Velocity Figures

From these fractional velocities, Brohamer derives several key figures:

- **Early Pace (EP):** The velocity to the second call (4f in sprints, 6f in routes). This is a measure of a horse's raw early speed.

- **Sustained Pace (SP):** A compounded rating of the second call velocity and the final fraction. This measures a horse's ability to carry its speed.
- **Average Pace (AP):** An average of all three fractional velocities. This gives a picture of the horse's overall pace profile.
- **Factor X:** A compounded rating that is particularly important in dirt sprints. It is a measure of a horse's ability to finish strongly after a fast early pace.

Source: [4]

4. Pillar 2: The Track Decision Model

Brohamer's second key insight is that **pace is not absolute; it is relative to the track.** A fast pace at Santa Anita is different from a fast pace at Belmont Park. The Track Decision Model is a framework for understanding the unique pace characteristics of each track, surface, and distance. [5]

This involves tracking recent races (the last 5-10 at that specific track/surface/distance) to determine:

- **What is the winning profile?** Do front-runners dominate? Do closers have a chance?
- **What are the average winning rankings?** For a 6f dirt sprint at this track, what is the average rank in Early Pace, Sustained Pace, etc., for winning horses?
- **What is the optimal Percent Early (%E) range?** (More on this in the next section).

This model allows you to move from generic pace analysis to a highly specific, contextual understanding of the race you are handicapping.

5. Pillar 3: Percent Early (%E) / Median Energy

This is one of Brohamer's most powerful concepts. It measures **how much of a horse's total energy is expended in the early part of the race.** [6]

The calculation is:

> ***%E = Early Pace (EP) / (Early Pace (EP) + Final Fraction (Fr3))***

(Note: Some sources use a simplified version, but the principle is the same. TrackMaster uses a proprietary version called “Median Energy” which represents the same concept). [4]

A high %E indicates a horse that uses a lot of energy early (a front-runner). A low %E indicates a horse that conserves energy for the stretch run (a closer).

The power of %E comes from comparing a horse’s individual %E to the **optimal %E range for that specific track and distance**, as determined by the Track Decision Model. A horse whose natural energy distribution fits the winning profile for that track is a strong contender.

6. Pillar 4: Turn Time (The Hidden Fraction)

Brohamer and the Sartin Methodology place enormous emphasis on the second fraction, the **Turn Time.** They call it the “hidden fraction” because it is not explicitly shown in standard past performances, but it is the single best indicator of a horse’s fitness and form. [3]

Quoting Brohamer directly:

> *“The ability to handle the turn in an efficient manner is a characteristic of a fit horse.” [3]*

Why is Turn Time so important?

- **It separates contenders from pretenders.** Any horse can run a fast first fraction. But only a fit, in-form horse can accelerate or maintain a high velocity around the turn.
- **It is a powerful form indicator.** A horse showing recent improvement in its Turn Time figures is a horse rounding into peak form.
- **It uncovers hidden ability.** A horse may have a modest final time, but a brilliant Turn Time, indicating it has more talent than its record suggests.

A word of caution: Turn Time is not a standalone factor. A deep closer may have a very fast Turn Time but be too far back to ever get into the race. It must be used in conjunction with the other pace figures and the Track Decision Model.

7. Synthesis: Putting It All Together

The Brohamer methodology is not a simple, mechanical system. It is a framework for thinking about a race in a new way. The process looks like this:

1. **Calculate the Velocity Figures:** Convert the times for all relevant past performances into feet-per-second for all three internal fractions.
2. **Consult the Track Decision Model:** Understand the winning pace profile for today's specific race.
3. **Analyze the Pace Shape:**
    - Who are the early speed horses (high EP, high %E)?
    - Will the pace be contested or uncontested?
    - Does the likely pace shape favor front-runners or closers, according to the Track Decision Model?
4. **Evaluate Each Horse:**
    - Does this horse's running style and energy distribution (%E) fit the likely race shape and the track's winning profile?
    - Is this horse in good form, as indicated by its recent Turn Time figures?
    - Does this horse have the necessary Sustained Pace or Factor X to finish the race strongly?

By answering these questions, you move beyond simply identifying the fastest horse and begin to identify the horse most likely to win *this specific race*, given its unique pace dynamics.

8. Conclusion: The Path Forward for TAT1

This framework provides the systematic approach to pace analysis that you identified as a critical gap. By integrating these concepts into your analytical process, you will be able to:

- **Predict race shape with greater accuracy.**
- **Identify horses whose running styles are advantaged or disadvantaged by the likely pace.**

- **Gain a deeper understanding of horse form and fitness through Turn Time analysis.**
- **Make more informed selections, especially in races where pace is the deciding factor.**

This is the bridge from the “science” of speed figures to the “art” of handicapping. It is the next logical step in your evolution.

References

[1] Brohamer, T. (1991). *Modern Pace Handicapping*. William Morrow & Co. [2] pacemakestherace.com. (2011). *Sartinian Pace Handicapping*. https://pacemakestherace.com/sartinian-pace-handicapping/ [3] Frandsen Publishing. (n.d.). *Pace Handicapping with Brohamer Figures, Part 4: “Turn-Time”*. Yumpu. https://www.yumpu.com/en/document/view/11495184/pace-handicapping-with-brohamer-figures-turn-time-brisnet [4] TrackMaster. (n.d.). *TrackMaster® MPH Pro Help Guide*. http://info.trackmaster.com/thoroughbred/mphhelp.htm [5] pacemakestherace.com. (2011). *Sartinian Pace Handicapping, Part 2*. https://pacemakestherace.com/2011/sartinian-pace-handicapping-part-2/ [6] HTR2. (n.d.). *HTR-Pedia*. http://htr2.com/km/pdf/HTR-PEDIA.pdf